a)

Process	Wait(full)	Wait(mutex)	Signal (mutex)	Signal (empty)
Initial Value	2	1		6
Consumer	**(a) 1**	**(b) 0**	**(c) 1**	**(d) 7**
Consumer	**(e) 0**	**(f) 0**	**(g) 1**	**(h) 8**
Process	Wait(empty)	Wait(mutex)	Signal (mutex)	Signal (full)
Producer	**(i) 7**	**(j) 0**	**(k) 1**	**(l) 1**

b) There are 2 concurrent processes wishing to enter their Critical Sections. Using Paterson's algorithm, complete Table 7 below (for items labelled as **a** – **o**). [9 marks]

Time	flag[0]	flag[1]	turn	Events
t_0	FALSE	FALSE	0	P0 executes RS0
t_1	FALSE	TRUE	**(a)0**	**(b)P1 request to enter CS1**
t_2	FALSE	TRUE	**(c)0**	**(d)P1 enter the CS1**
t_3	**(e)TRUE**	TRUE	1	**(f)P0 request to enter CS0**
t_4	TRUE	TRUE	**(g)1**	**(h)P0 is busy waiting**
t_5	TRUE	FALSE	**(i)1**	**(j)P1 execute RS1/ P1 exit from CS1**
t_6	TRUE	FALSE	**(k)1**	**(l)P0 enter/execute CS0**
t_7	FALSE	FALSE	**(m)1**	**(n)P0 execute RS0/P0 exit from CS0**
t_8	FALSE	TRUE	**(o)0**	P1 requests to enter CS1